



The LED Lamp metaphor: Knowledge and the creative process in new media art[☆]

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ABSTRACT

In the context of new media art, the intrinsic interplay of disciplines poses constant learning challenges to the artists, as it requires a continuous acquisition and reconfiguration of knowledge. This article seeks to understand to what extent knowledge processes of creation, transfer and adoption contribute to the creative collaborative outcomes of new media artists, by investigating: i) explicit and tacit knowledge flows of new media artists; ii) strategies for knowledge creation, transfer and adoption; iii) contexts in which these processes are facilitated. Based on qualitative methods, findings indicate that the way in which artists learn and interact is related to their personality; values such as openness, curiosity, respect or trust shape the context for a shared interest in knowledge creation, adoption, and exchange. A holistic approach to knowledge-creating place ('Ba') contributes to understanding the fluid contexts in which creative processes occur.

1. Introduction

Knowledge has been recognized as playing a major role in creativity as its acquisition and recombination support the generation of novel ideas, which in turn foster creative thinking and problem-solving (Amabile, 1998; Amabile et al., 2018; Mumford, Giorgini, Gibson & Mecca, 2013; Scott, 2010). Fields of high uncertainty, such as the cultural and creative sectors, rely on social networks (Potts, Cunningham, Hartley & Ormerod, 2008; Scott, 2010; van Dijk, 2014) and use collaborations to access and share knowledge (Bathelt, Malmberg & Maskell, 2004; Currid, 2007; Scott, 2011; Wenting, Atzema & Frenken, 2010). These collaborations often occur in informal settings (Clare, 2013), in a combination of virtual and physical presence and exchanges. Technology plays a role in these processes as a means of communication, but also for expanding creative possibilities, in particular regarding technological-related knowledge (Cox, Haskel, Poulter, Stevens & Wallbank, 2004; Powell, 1998; Wasko & Faraj, 2005). It is therefore not surprising that knowledge-based economies are founded on the valorization of knowledge and technology for the productivity and economic

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performance of societies (Dubina, Carayannis & Campbell, 2012; OECD, 1996).

Within the creative industries, new media art emerged during the late 20th century at the intersection between art, science, and technology (Gere, 2008). New media art can be characterized by the use of technology as a medium (Paul, 2008), as well as a reflective approach to the social, political, and cultural implications of such media (Brouwer, Fauconnier, Mulder & Nigten, 2005; Quaranta, 2013). The intrinsic interplay between artistic and scientific disciplines poses constant learning challenges to the artists, as it requires a continuous acquisition and reconfiguration of both explicit and implicit knowledge (Scott, 2010). These processes are also part of the discussion as problematised by Ross (2010) when discussing the idea of expertise in new media production. Despite this and the fact that literature on knowledge management within organizations is diverse, there is scant information on the processes, flows, strategies and contexts where knowledge is created, adopted and transferred among (new media) artists.

The present article aims at investigating the knowledge arena of (new media) artistic collaborations, examining the interplay between explicit and tacit knowledge beyond its divide and proposing a holistic understanding of the creative process within the framework of 'Ba', a knowledge-creating place (Nonaka & Toyama, 2003). This research seeks therefore to understand to what extent knowledge processes – creation, transfer and adoption – contribute to the creative collaborative outcomes of new media artists. More specifically, it investigates: i) (explicit and tacit) knowledge flows of new media artists; ii) strategies for knowledge creation, transfer and adoption; iii) contexts in which knowledge creation, adoption and transfer are facilitated in the creative process. In order to delve into these issues, a qualitative approach is taken, examining contexts where little research on new media art has been done, such as in Mexico City (Villagomez, 2017), where new media artworks are emerging amid a number of challenges.

2. New media art, a dynamic hybrid genre

The origins of what is now called new media art are rooted in the technological advances enabling data processing and storage during and after the Second World War (Gere, 2008). Media art emerged and developed in the 1970's. The term new media art surfaces in the mid 1990's, referring to the convergence of the mass media, and the new capabilities of data processing and its intersection with art (Quaranta, 2013). However, the concept of new media art is not consensual (Kimbell, 2004; Quaranta, 2013), which would correspond to the aesthetics of hybrid digital media (Paul, 2008). Other terms have been used to refer to the same type of artworks such as media art, digital art, and electronic art; or, even more specifically, net art, software art or robot art. Most of these are obsolete (Quaranta, 2013), which explains why both media art and new media art the most widely used terms to refer to these artworks. Besides, despite its limitations, one of the best qualities of the genre is the freedom that arises from a lack of a delimited definition (Paul, 2008), as the constant evolution of digital technologies opens up new possibilities for the creation and the experience of art.

New media art is therefore an artistic genre spanning several disciplines, including, but not limited to, visual and performing arts. In these artistic practices, art, science, and technology are inherently intertwined. Although other art forms nowadays often use technology, new media art uses science and technology as a medium, instead of as a tool (Paul, 2008). New media art uses computational algorithms to produce, store, and present the artworks (Paul, 2008). The exploration with digital technologies in this field is characterized by the use and development of new applications, and also by their distortion and reconfiguration (Brouwer et al., 2005). New media artworks are usually dynamic, process-oriented and time-based; and may be participatory, interactive, immersive, collaborative, performative, modular, variable, generative, customizable, and networked (Gonsalves & Chan, 2013; Paul, 2008).

The production of new media artworks, therefore, involves the use of a wide range of knowledge and inputs from diverse fields, mainly science, art, technology, and design. These inputs are often gathered, researched and developed through collaboration between interdisciplinary teams of artists, designers, technicians and scientists (Brouwer et al., 2005). New media technologies have a strong impact in society and in artists' practices. As portrayed by Silverstone (1999), new technologies 'do new things. They give us new powers. They create new consequences for us as human beings. They bend minds. They transform institutions. They liberate. They oppress' (p. 10). The role of artistic research in this field extends beyond making use of available technology or creating new technological configurations, into reflection on the implications and connotations of technology in our society, a process in which both tacit and explicit knowledge are in a constant interplay. Being an ever-evolving field, new media art offers many (knowledge-related) challenges to artists who rely on collaborative networks to build conceptual and technical bridges of understanding (Druckrey, 2005).

3. Knowledge in cultural and creative production

In the production of cultural and creative activities, human capital is considered the main resource for novelty generation and product differentiation (Scott, 2010), representing a valuable competitive advantage in uncertain environments (Wasko & Faraj, 2005). In economic terms, the performance of individuals and organizations has become dependent on their ability to create and use knowledge (Phelps, Heidl & Wadhwa, 2012). Beyond the impact on economic performance, knowledge plays an important role in many human activities since it is closely related to creativity, cultural capital and novelty generation (Bourdieu, 1984; Lizardo, 2017; Mumford et al., 2013; Scott, 2010). Research on creativity suggests that it is most valuably understood as a process of three phases: idea generation, idea promotion and idea implementation (Caniëls, De Stobbeleir & De Clippeleer, 2014). The acquisition of relevant knowledge supports the creation of novel ideas (Accominotti, 2009; Mumford et al., 2013; Scott, 2010; van Dijk, 2014) which can be helpful in all of the aforementioned stages. Creative thinking, therefore, originates with the knowledge and skills available through education, experience, and informal social exchanges (Scott, 2010), which can be dependent on context and lead to different forms of social, cultural and symbolic capital (Bourdieu, 1993).

The literature on knowledge and knowledge management often makes a distinction between different types of knowledge, although in practice, forms of knowledge are intertwined. In general terms, knowledge is about intangible assets, operational routines, and

creative processes in the form of new information, expertise, and ideas (Wasko & Faraj, 2005). On the one hand, explicit knowledge consists of articulated, structured, information that can be gathered and expressed through information technology (Ajmal & Koskinen, 2008; Saint-Onge, 1996) and includes all the information that is consciously expressed after its rationalization (Nonaka & Toyama, 2003). On the other hand, tacit (or implicit) knowledge is the unarticulated information gathered through perception and experiences in the form of intuition, perspectives, beliefs, and values (Ajmal & Koskinen, 2008; Saint-Onge, 1996), and it is often not rationalized (Nonaka & Toyama, 2003).

In an attempt to conceptualize culture as knowledge, Lizardo (2017) reflects on the implicit / explicit divide by exploring it from the perspective of cultural analysis. Drawing on the work by Strauss and Quinn (1997) on declarative and nondeclarative culture, Lizardo (2017) adds a contextual factor to the discussion, an element only marginally present in Bourdieu's approach, as 'Bourdieu's ambivalence in relation to the culture concept essentially meant he made little effort to provide an empirically defensible account of how persons acquire and use culture in context, outside of some suggestive, but ultimately elliptical, allusions' (p. 89). As declarative culture can be linguistically articulated, much is therefore left to interpreted meanings in this non-declarative form of culture, one that is 'phenomenologically opaque and not open to linguistic articulation' (2017, p. 89). Context plays a role in the encoding and decoding processes (Hall, 1980; Lizardo, 2017) involved in knowledge creation, transfer and adoption which pervade the cultural sphere in different ways.

The typology developed by the OECD (1996) – know-what, know-why, know-how and know-who – portrays distinct focus points of knowledge processes. The first two types, know-what, know-why, correspond to articulated, declarative or explicit knowledge, whereas the last two, know-how and know-who, refer to unarticulated, nondeclarative or tacit knowledge. Both levels of knowledge are highly valuable, as they become important resources for daily performance. However, it is in the interaction and flows between tacit and explicit knowledge that new knowledge is created (Ajmal & Koskinen, 2008; Nonaka & Toyama, 2003) (see Table 1). As pointed out by Dhanaraj, Lyles, Steensma and Tihanyi (2004, p.430), 'whereas explicit knowledge provides the building blocks, tacit knowledge provides the glue and integrating mechanism in learning' and the acquisition of both can happen simultaneously.

Much of the acquisition of tacit knowledge occurs in the informal interaction with others during which individuals generate, access, and integrate new knowledge. Tacit knowledge can only be transmitted through an active involvement of the party providing it. However, it does not require standardized processes, it is not easily codified and cannot be easily reverse engineered (Dhanaraj et al., 2004). In contrast, explicit knowledge is easily codified and transferred, even without the active involvement of its source (Dhanaraj et al., 2004).

Human interaction has been widely recognized as an important learning mechanism: through communication and socializing, humans are able to detect the assumptions of others, recognize their own assumptions, and understand how that affects their reactions (Bathelt et al., 2004; Nonaka & Toyama, 2003; Saint-Onge, 1996). Social networks therefore represent a driving force for knowledge exchanges, and particularly knowledge creation, which often arises from the interaction between different entities (Nieves & Osorio, 2013; Phelps et al., 2012).

Knowledge networks are formed by the social relationships of individuals, collectives, and organizations, who heterogeneously possess knowledge, and who, through their interactions, can search, create, transfer, acquire, and use knowledge (Phelps et al., 2012; Powell, 1998). Therefore important social processes such as sharing, interpreting, and combining information are important in knowledge exchanges or transfers (Argote, Ingram, Levine & Moreland, 2000).

The importance of these interaction processes in different spheres of society have been conceptualised by (Collins, 2004), inspired by the work of Goffman. (Collins, 2004) interaction ritual chains provide a framework to understand knowledge processes and in particular knowledge creation as the result of collective contexts where emotional energy is generated. This resonates with Becker's view of art worlds, where 'various groups and subgroups share knowledge of the conventions current in a medium, having acquired that knowledge in various ways. Those who share such knowledge can, when the occasion demands or permits, act together in ways that are part of the cooperative web of activity making that world possible and characterizing its existence' (1982, p. 67). It is therefore

Table 1
Tacit and explicit knowledge

Concept	Description	Authors
Tacit knowledge	Unarticulated information gathered through perception and experiences: intuition, perspectives, beliefs, and values. It is useful to filter, interpret, and understand experiences which will affect decision-making, behavior, and performance.	(Ajmal & Koskinen, 2008; Saint-Onge, 1996)
	Only transmitted through an active involvement of the one providing it; it does not require standardized processes.	(Dhanaraj et al., 2004)
	Information is often not rationalized.	(Nonaka & Toyama, 2003)
	Nondeclarative (cultural) knowledge: phenomenologically dense, not always clear and it cannot be linguistically articulated. It is related to 'knowledge how'.	(Lizardo, 2017)
	Know-how: the skills to do something; Know-who: information of social relationships that allows the connection with others who possess knowledge and use it meaningfully.	(OECD, 1996)
Explicit knowledge	Articulated, structured, information that can be easily gathered and expressed through information technology.	(Ajmal & Koskinen, 2008; Saint-Onge, 1996)
	Information is easily expressible after rationalization.	(Nonaka & Toyama, 2003)
	Declarative (cultural) knowledge: phenomenologically transparent, it is linguistically articulated. It is related to 'knowledge that'.	Lizardo, 2017
	Know-what: knowledge about facts and information; Know-why: scientific knowledge of the principles and laws of nature.	(OECD, 1996)

common for knowledge-based festivals to use networks at different levels (Podestà & Richards, 2018), generating complex contexts which require a 'multi-layered interpretation of the social interaction processes' (Marques, Borba & Michael, 2021, p. 21).

Knowledge exchanges occur in a specific context which also influences the processes and people involved. Arguably, the interplay between different types of knowledge and contexts is what makes knowledge processes dynamic and prolific, in particular in the creative sphere. In this respect, the Japanese philosophical concept of Ba (Nonaka & Konno, 1998; Nonaka & Toyama, 2003) provides an interesting approach and a fertile ground for conceptualizing knowledge complexities in creative processes. Ba is used specifically within the context of organizations to describe a shared place or context (either physical or virtual) in which knowledge is exchanged between its participants through their interaction. This knowledge-creating place (Ba) allows for contradictions to arise and create new knowledge. This creative driving force is fundamental, we argue, to the arts and culture, as the intention to share and compare ideas, perceptions and information is what creates the Ba. This perspective stems from Nonaka and Takeuchi's (1995) SECI model of knowledge dimensions in organisations, in which the interaction between explicit and tacit knowledge results in four types of conversion: Socialization (from tacit knowledge to tacit knowledge); Externalization (from tacit knowledge to explicit knowledge); Combination (from explicit knowledge to explicit knowledge); and Internalization (from explicit knowledge to tacit knowledge). The interaction between humans and their environment starts the process of knowledge creation and continues with an ever-growing process of socialization, externalization, combination, and internalization of tacit and explicit knowledge (Nonaka & Toyama, 2003). These interactions and interplay of different forms of knowledge occur in a similar way in the artistic realm.

Criticism of the SECI model and the term 'tacit knowledge' has also been voiced (e.g. Ray, 2009; Tsoukas, 2003), focusing, for example, on the idea that the tacit dimension (Polanyi, 1983) of knowledge can be transferred into the explicit sphere (Nonaka & Toyama, 2003). Ray (2009) argues that Nonaka and Toyama's (2003) approach omits a fundamental part of the tacit dimension, which relates to the unsaid, to faith, and metaphysical truth.

One can argue however that the concept of Ba provides the metaphysical aspect that Ray (2009) considers to have been ignored. This knowledge-creating place provides a platform for knowledge conversion, which can also be seen as including unsaid elements. As Nonaka and Toyama's (2003) account of knowledge-creating place indicates, research on knowledge and organizations draws attention to certain relational characteristics such as interpersonal trust, commitment, reciprocity, respect, and openness to other ideas (e.g. Kessel, Kratzer & Schultz, 2012; Wasko & Faraj, 2005; Zakaria, Amelinckx & Wilemon, 2004). Such relational traits are more often nourished in close relationships, which therefore become particularly relevant in virtual spaces of collaboration, characterized by an absence of face-to-face interaction. Building strong bonds and team dynamics is important to facilitate knowledge sharing and achieve the group's objectives effectively (Wasko & Faraj, 2005; Zakaria et al., 2004), which is also in line with Collins' (2004) idea of mutual focus. In 1999, Robins had already emphasized the need for 'meaningful politics of engagement' (p. 24) to develop relationships that can yield knowledge exchanges in virtual spaces.

To better understand knowledge flows and processes, Phelps et al. (2012) suggest looking into knowledge outcomes. Three conceptual types are suggested: 1) knowledge creation – new ideas, practices, products or services; 2) knowledge transfer – diffusion of knowledge between actors; 3) knowledge adoption – integration and use of acquired knowledge, which involves the decision to do so. These outcomes are closely related to each other in the natural process of human cognition.

Another complementary perspective, based less on the type of knowledge and more on the knowledge objective, is provided by Snider and Nissen (2003). In this case, knowledge flows can be thought of as 'solution', referring to the real-time transfer of knowledge to solve problems in a particular setting; as 'experience', referring to the accumulation of knowledge for future use; and as 'socially created', referring to the knowledge that is created and shared among individuals in social relationships (Ajmal & Koskinen, 2008; Snider & Nissen, 2003). This last category points to the importance of informal social interactions and communications, embracing discussion, argumentation, negotiation, and decision that foster the creation and transfer of knowledge (Nonaka & Toyama, 2003; Phelps et al., 2012).

Collaborative efforts commonly take the form of projects, a central way of organizing cultural and creative activities. Projects are often unique, uncertain, and complex (Ajmal & Koskinen, 2008) bringing together a pool of people with different skills and knowledge to realize a set of objectives. Furthermore, they are often constrained in resources, such as time and budget. In these forms of organization, the effectiveness of the transfer of knowledge varies, depending on the capabilities of the team members to create, value, absorb, and share knowledge (Ajmal & Koskinen, 2008). However, the creative process is not only affected by the group composition (Amabile, 1998) but also by their individual mindsets, the rewards perceived, the sense of leadership (or lack thereof) and the resources they have (Caniels et al., 2014), all of which rely on a combination of tacit and explicit knowledge of the participants. In professional collaborations, strong social ties can be identified by the extent to which the parties commit themselves to the exchange of emotional support, managerial expertise, and interaction time between them (Dhanaraj et al., 2004). Collaboration as well as inspiration and knowledge are key themes in the conceptualisation-production interactions that result in art creations (Accominotti, 2009; Lehman, Wickham & Fillis, 2018; van Dijk, 2014).

In high uncertainty environments learning through interaction with other individuals and organizations is an important source of skills, ideas, resources and expertise, or in general terms, a source of tacit and explicit knowledge (Powell, 1998). Likewise, technology-related fields are also drawn to the use of collaborations to access new knowledge (Cox et al., 2004; Powell, 1998; Wasko & Faraj, 2005). This is particularly relevant for the context of new media art, the set of artistic practices that involve the use of technology and science. However, if literature on knowledge management is abundant, little is known about the knowledge processes and flows of independent artists, in particular of new media artists, which can contribute to better understand their strategies for knowledge creation, adoption and transfer and in what contexts these can be facilitated.

4. Methodology

4.1. Empirical setting: new media art landscape in Mexico City

This study is focused on the new media art landscape in Mexico City due to its socio-artistic specificities and the fact that research on new media art is still scarce in this region of the world (Villagómez, 2017). As the main city in Mexico, it is not surprising that Mexico City has been leading the conversation with regard to art and technology in the country, supported by the infrastructure of academic and artistic institutions. The creation, exhibition, and documentation of new media art in Mexico City is a collaborative effort between different actors, namely artists, scientists, public institutions, and art organizations. Moreover, Mexico City hosts most of the exhibition spaces for new media art in the country (Villagómez, 2017). The most famous are the Centro Multimedia, Centro de Cultura Digital, Laboratorio Arte Alameda, and Museo Ex Teresa Arte Actual, all of which are owned by the Mexican government. Nevertheless, new media art has found exhibition and research spaces in many independent settings, both online and offline.

Although Mexico City is a place where the creation and exhibition of new media art is thriving, very little research exists (Villagómez, 2017). Understanding the context of knowledge processes and flows in Mexico City is particularly important because Mexico has the least knowledge-based economy among the OECD countries (OECD, 2017). The OECD therefore sees improving skills and knowledge flows in Mexican society as key future priorities. Thus, exploring the learning processes of a dynamic and interdisciplinary practice, where collaboration is important, can help to identify new ways in which knowledge and creativity can be fostered in the country and also be relevant for other places around the globe.

4.2. Research design and methods

The present research uses an exploratory qualitative approach which allows us to unearth processes and understand flows by capturing important nuances in the creation and reflection of knowledge, particularly, tacit knowledge. In this manner, the artists' attitudes, values, opinions and their interpretation of experiences, events, and understandings (Byrne, 2012) could be studied in relation to knowledge exchanges. A total of fourteen in-depth, semi-structured interviews with new media artists, with estimated ages ranging between 28 and 45 years old, were conducted between 23 April and 13 May 2018 via online video or audio-conference. The artists were asked about their artistic practices, social networks, collaborations and projects, levels of knowledge (explicit and tacit), as well as creation and transfer of knowledge. The sample consists of people who use science and technology as media to produce, store and present their artworks and reflect on the social, political and cultural implications of their use, following the description of the field used in this study (Brouwer et al., 2005; Paul, 2008; Quaranta, 2013). Since the boundaries of new media art are blurred and the artists often move through and across different digital media and technologies, no specific requirements were established in terms of media used. All artists had an active artistic practice and their work has been well received and recognised, which is supported by noticeable achievements such as exhibitions, grants received, and active entrepreneurial projects and job positions in cultural institutions in the country and abroad. Moreover, all artists in the sample have a strong bond to the city, regardless of their nationality. A snowball sampling method was used, as it proved to be the way to access the artists' networks. Different contacts were used as the starting point of the snowball to try and reduce any possible bias. All interviews were recorded and transcribed in the original language (Spanish) and quotes used in this paper were translated by the authors. The data were analysed thematically. This consisted in the creation of codes, sub-categories, and categories, assembled in themes. The coding technique was mixed, as it included codes that were decided a priori based on the literature, and, after reading the interviews, emergent codes were integrated.

As with any method, interviewing has its advantages and disadvantages. Related to the latter, and a limitation to this study, is the fact that access to tacit knowledge is a challenge, in particular when there is an attempt to put it into verbalised forms, i.e. making them explicit. Being an exploratory study, this research attempts to: i) approach the interplay of explicit and tacit knowledge as perceived and told by the artists themselves; ii) consider tacit knowledge in terms of what has not been said as well as interpretations of the silenced and hidden aspects beyond the artists descriptive accounts. In this sense, tacit knowledge was observed and analysed when the interviewees referred to ideas, perspectives, beliefs, values, as well as social relationships and non-explicit aspects of their know-how. Explicit knowledge was observed and analysed when particular technologies, facts, documents, books, and specific technical information were mentioned.

5. New media artists in Mexico City

The educational and practice-based backgrounds of the interviewees differ significantly. Most of them have an artistic educational formation, sometimes not completed. However, they all converge in the sense that they come from disciplines that involve creativity and the constant flow of ideas in their learning processes. The background disciplines are varied: philosophy, art, music composition, visual design and multimedia communication, architecture, and industrial design. More importantly, they all have moved through and across different disciplines throughout their artistic practice. The main common characteristic is that they have developed skills in digital technologies, mainly in the use of computer languages.

Throughout their artistic career, the interviewees have worked with many different media, which makes it hard to categorize their artistic practices. They have used diverse technologies to explore esthetic and ideological inquiries. Only those who started their artistic career more recently seem to be more specific about the type of media they use. Nevertheless, their work is mainly situated at the intersection of art, science, and technology. For this reason, and due to the mixed type of used means in the making of their works, these artists would fall under the new media art label, which is in itself non-consensual and very broad (Kimbell, 2004; Quaranta,

2013).

It becomes clear from the findings that interdisciplinarity, as [Brouwer et al. \(2005\)](#) and [Fawcett \(2013\)](#) pointed out, is essential as a concept to understand the practices, processes, philosophies, and work of new media artists. Thus, it is not surprising that the most predominant description of the artistic practice of the new media artists interviewed is the reflective use of technology and the use of interdisciplinary collaborative efforts that come together in an artistic work:

In general, I believe that this is the future not only for the arts, but for all sciences. It is not just that art can learn from biology or other sciences, it is also science and technology which can learn from the methods of artistic production. (...) It is an important way of working because one needs many perceptions about the world and when one only has one it is easier not to be creative. In the end, being creative is to combine ideas. (Interviewee 11)

This interdisciplinarity is not related to individuals, but is rather a result of collaborative efforts, which complement, give feedback or retro-feed individual knowledge. In new media art production, findings suggest that the focus is much less on the outcomes and much more on the process, as [Paul \(2008\)](#) had previously indicated. Different forms of knowledge and strategies to create, access and transfer it are the backbone of this liquid art form ([Bauman, 2007](#)). The main reasons for new media artists to exchange knowledge are, according to the findings, based on their willingness to transfer and acquire knowledge from different disciplines. They are highly driven by the curiosity to learn from other fields, particularly when they are looking for ways to develop their ideas, but lack the knowledge to fully implement them. The paramount perception of artists is that new knowledge helps them develop new ideas and perceptions that inspire their creativity. These exchanges among artists seemed to be motivated in the first instance by the vision of a shared interest ([Collins, 2004](#)), for which some personality traits set the background and work towards facilitating that process. In the context of new media artists in Mexico City, findings suggest that the cultural homophily mechanism stimulates ties between individuals who do not necessarily occupy the same position in the network, but, in fact, might belong to contesting fields ([Basov, 2020](#)). Very often, the collaboration is with artists from different background disciplines, with a distinct range of skills and expertise. However, it is clearly suggested that the sphere of negotiations of knowledge of new media art goes beyond the figure of the 'artist', as there are participants in this exchange process who might not necessarily identify as artists. The 'artist' label is not the most important, as in these collaborations labels become fluid as the artistic idea transcends the notion of art versus non-art. It is in the process that the new media art world configures itself and gains its characteristic existence ([Becker, 1982](#)), which can be seen as a LED lamp (see below), where the importance lies less in the end result and more in the creative process.

6. The LED lamp metaphor or knowledge in the creative process

The transfer and acquisition of new knowledge enables the expansion of cognitive abilities and skills, especially when these processes involve tacit knowledge in the form of ideas or perceptions, as previously pointed out by [Dhanaraj et al. \(2004\)](#). Besides, most of the new media artists interviewed expressed their willingness to share the knowledge they have with other people in different ways, from direct collaboration in a specific project to sharing information online available to the public:

We're not afraid of sharing the process of how to do things (...). [In this specific form of art] we artists are not afraid of sharing the codes, sharing the knowledge, sharing the way of creating, because it's the internet after all. (...) [What is important is] that people share the content you create. (Interviewee 8)

Most of the artists were very positive about sharing, which contrasts with some others, who valued their knowledge and were somehow more protective. Many of these artists also assume an active role in passing on knowledge (knowledge transfer) in formal settings such as workshops, courses and other educational programs, where they take the role of knowledge facilitators. These educational programs most frequently happen within institutional formal contexts and involve the use of explicit knowledge, with information being explicitly articulated and structured, and expressed with awareness of its existence after its rationalization ([Ajmal & Koskinen, 2008](#); [Nonaka & Toyama, 2003](#); [Saint-Onge, 1996](#)).

Being involved in knowledge exchanges ultimately affects the interviewees' artistic development. However, the process of changing the relationship from mere contact with someone to the development of knowledge exchanges is not always driven by a clear intentionality, since other factors come into play. It is mostly done in an 'organic' way, as several interviewees mentioned:

(...) somebody invites you to collaborate in a project that they are doing, (...) to be part of the activity (...) But I believe that the most organic and comprehensive way is when certain bonds and chats are established, then you can share discussions, or share dialogues (...) And I insist, this happens in a very organic manner. (Interviewee 10)

I think the collaboration has to be very organic, that is, if you put it in words and plan, it doesn't work. (Interviewee 13)

This organic and often informal development of a collaborative relationship, which often develops beyond the professional context (friendships, intimate relationships), is different from what happens in more formal professional contexts. The literature on knowledge management is in fact focused on (project-based) organizations ([Ajmal & Koskinen, 2008](#); [Nonaka & Toyama, 2003](#); [Phelps et al., 2012](#); [Snider & Nissen, 2003](#)). The theories about knowledge management in organizations or projects can be useful to better understand how knowledge processes and flows work in these collaborative contexts. It is important to bear in mind that despite the common projects or even artistic collectives (a more structured form of organization), these media artists acts as individual artists and it is in that capacity that they consider their artistic activities. This makes it more challenging to map knowledge flows, as well as the creation, adoption and transfer of knowledge.

The artists' motivations rely on shared interests at a specific moment in time, in alignment with knowledge-creating place ([Nonaka & Konno, 1998](#); [Nonaka & Toyama, 2003](#)), which can be a trigger for the project that will eventually lead to a collaborative creative



Fig. 1. Flujo [Nowhere/NowHere] (2015) A Triple (Fermín Martínez + Tonalli Villalpando).

outcome:

In that horizontality [interdisciplinary collaboration] where there is no power struggle, but rather a common interest of crossing the other's knowledge. I think it's like a vulnerability space. And these vulnerability spaces which recognize the knowledge of the Other and the Other recognizes my knowledge, and we both recognize a mutual interest. (Interviewee 12)

This example illustrates how tacit and explicit knowledge are combined, which is very precisely illustrated by the metaphor of an LED lamp:

In the end, all the collective effort may be done to turn one LED lamp on. Though the important thing is not if the LED is turned on at the end of the experience, but what you can comprehend or experience in the process. (Interviewee 7)

More than one artist in the study used the metaphor of an LED lamp to describe the importance of the process in new media art. The idea behind the metaphor relates to the final apparent result of the lamp turning on being less important than the process of creating the artwork itself, as well as the dialog the artwork intends to provoke: 'It was very easy to [...] open this type of [new media art] experiences to a diverse audience who are interested in this critical dialog. [...] [There is an] overlap of interests' (Interviewee 10). The focus is on the process, guided by participatory unspoken rules, clearly related to the creation, adoption and transfer of tacit knowledge. In this process, informal collaborations with other artists evolve into more focused creative endeavours (see Fig. 1). As one of the interviewees expressed: 'At some point we became sort of a collective, where even if we were not producing [an artwork based on] the same idea at the same time, there was a constant network of exchange, of flow, of learning and sharing' (Interviewee 12). This artist explained how, in the collaborations with this group, and particularly with one dancer, they started testing sound devices that were initially developed for other projects. Through this creative interaction, the interviewee felt the need to engage with movement themselves – an activity foreign to their regular artistic practice in an attempt to explain or to understand what the others meant. The artist's research on movement, sound, and the body continued well after these first encounters, even if:

At first sight, it might look like it's about things which are not very precise. But that's exactly what is so cool about it (...) That is when, I think, it happens, when words cannot say it, that is when it is not really precise. (Interviewee 12)

Everyone involved has to acknowledge and accept these rules (part of the ritual ingredients, Collins, 2004), in order for the design and, ultimately, outcomes to be possible. In this context 'artistic research is distinguished from scientific and technological research by the fact that it is not a means of reflection and theory formation (like scientific research) nor problem-solving and product development (like technological research), but is itself a form of reflection' (Brouwer et al., 2005, p. 7). These constant transfers contribute to the artists' creative identity (Scott, 2012) at the crossroads of different disciplines, which are inclusive, or even transversal.

7. Knowledge processes: creation, transfer, and adoption of tacit and explicit knowledge

The interviews show that new media artists are able to create, transfer, and adopt knowledge from the interactions they have with people that belong to their social networks, albeit the fact they might be in contesting fields (Basov, 2020). These knowledge processes are closely interrelated (Phelps et al., 2012) in different ways. Knowledge transfers in these collaborations occurring in formal and informal contexts are reciprocal and include tacit and explicit knowledge, which, when combined, create new knowledge, which, in turn, retro-nurtures knowledge transfer and adoption, feeding back into the cycle, as shown in Fig. 2.

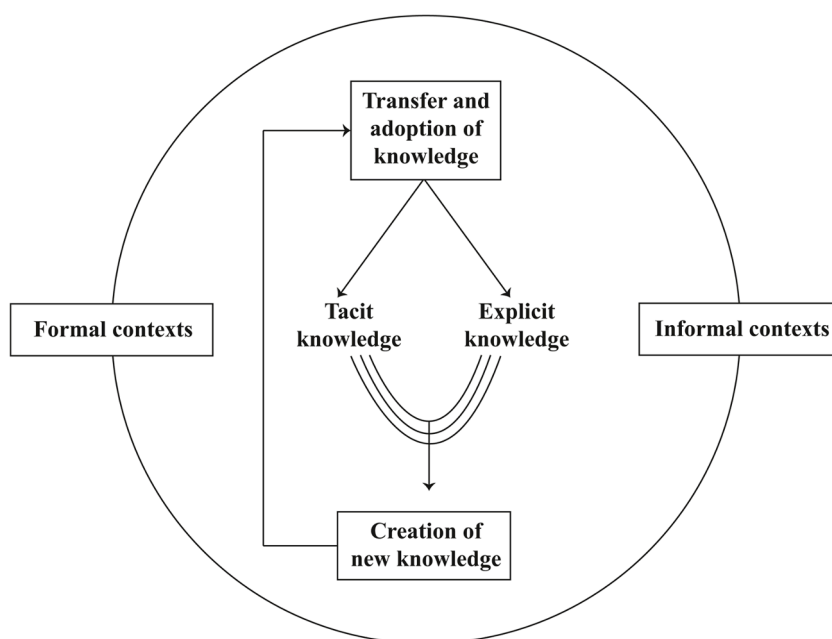


Fig. 2. Knowledge processes of tacit and explicit knowledge.

In these processes, the end result, e.g. the LED lamp, is less fundamental than the exchanges themselves. These ideas of feedback, retro-nurturing, and complementarity are described by the interviewees in different ways, very often in the form of knowledge within different types of collaboration being created and circulated. As interviewee 6 pinpoints: 'It's always cool, this feedback between all of us and the exchange of information and knowledge that we have; it's open'. This is also related to the artists' sharing philosophies and practices, as it is seen as a way to not only be grateful, give back, but also to build on and add to extant knowledge. However, it becomes clear that there is a pattern for these new artists to share a certain (tacit) ethos about the knowledge they are involved in creating.

In fact, the findings suggest that new media artists value tacit knowledge more than explicit knowledge, which becomes apparent in their exchanges. Having conversations about the ideas, experiences, and perceptions that spring from a certain technology is more emphasized than technical discussions. Nonetheless, explicit and tacit knowledge are not often conceived of separately since both types of structured and unstructured information arise in conversations and in formal / informal learning moments. Learning is effectively another important trait in these knowledge processes. Some artists emphasized their self-learning skills, particularly important because the acquisition of technical knowledge poses a constant challenge. New media artists seem to be quite aware of the constant and rapid change in digital technologies:

We give a higher importance to conceptual knowledge than to technical knowledge. This is because one of the lessons we have had in working with new media is that there is great obsolescence in technical knowledge. If you base your practice on the latest technologies, you are always going to be chasing behind it. (Interviewee 10)

Moreover, all the artists recognized and emphasized that constant research is fundamental to their artistic practice. They gather knowledge from communication channels that provide structured information, i.e. explicit knowledge, such as video tutorials, manuals, non-fiction books, and other resources which they commonly find online. This type of search is often connected to looking for a solution for a specific issue (knowledge as a solution, Snider & Nissen, 2003). Other sources of research were also identified, particularly in search of inspiration for creative endeavours. Artists often mentioned that one of the main sources of ideas, i.e. tacit knowledge, is found in other creative contents such as fictional books, movies, music, and artistic performances. Both sources feed different types of knowledge flows (see Fig. 3), explicit knowledge being more connected to knowledge as a solution, while tacit knowledge being more present in knowledge as experience and, mostly, socially created.

The findings suggest that the most frequent type of tacit knowledge that artists receive and share through interaction with others are further social relationships, or access to social networks, labelled by the OECD (1996) as 'know-who'. This is also consistent with previous research, for example, on artistic communities and collectives (Accominotti, 2009; van Dijk, 2014). The importance of social networks was identified in every interview, although it was rarely explicitly expressed as knowledge. The most frequent type of tacit knowledge artists gain from others are different perspectives, as highlighted by some interviewees who state that 'having a good idea is fundamental (...). Arriving at a good idea, in any context, not only in art, becomes easier and more natural with an external input, through sharing of ideas' (Interviewee 4). Ideas are also outcomes of knowledge creation in collaborative processes. In this context, 'it is a type of knowledge that is always in movement. (...) I could say that none of the ideas I have developed are mine, all of them have sprung through contact with other people' (Interviewee 11). Social interactions are sources of inspiration and ideas exchange. Knowledge in these new media art milieus are socially created, and foster creativity and novelty. This involves learning also at an

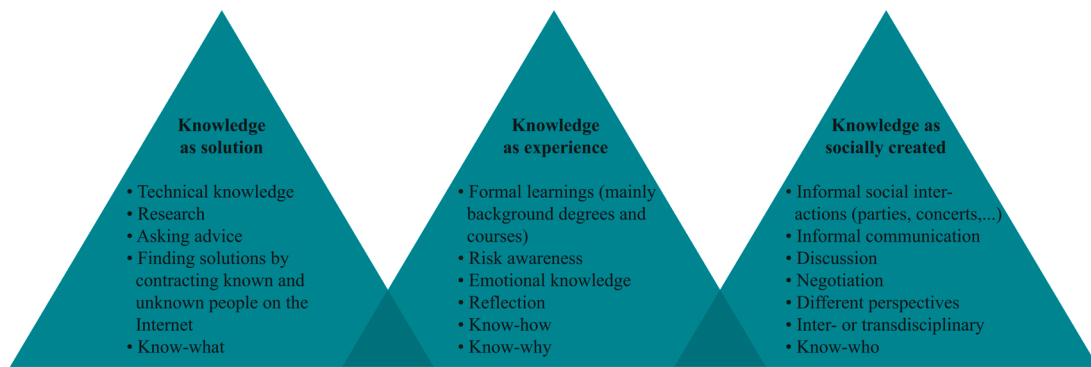


Fig. 3. Knowledge flows.

emotional level, which allows the artists to analyze situations and learn how to react, particularly in difficult situations (Bathelt et al., 2004; Saint-Onge, 1996). And if it is evident that personality defines this emotional knowledge, in the findings, certain characteristics and shared values which act in different types of knowledge contexts become apparent:

I was working in a scientific visualization lab and I was asked to take a course on inter- and transdisciplinary research. There I met a guy that told me about his tutor, who also worked with visual art programming and he introduced us. We were working on a similar project so I showed him my code and he showed me his... it was 'love at first code'. Then we continued working together. (Interviewee 9)

8. Values of knowledge-creating place for new media artists

Socialization skills are of great importance for the people in the art world to foster their careers with other artists and gatekeepers (Currid, 2007). Even the ones that mentioned preferring working alone or only virtually, acknowledge that opportunities for socialization allow the access to different disciplines where other behaviours and values are shared. Establishing a dialog not only fosters the access to networks and social relationships, in itself a form of knowledge (OECD, 1996), but it was identified as the most valuable communication channel to share knowledge. In these knowledge exchanges, the relational embeddedness between two actors, i.e. social ties in a relationship and a mutual feeling of trust, play major roles (Scott, A.J., 2010, 2012). Strong relationships are further created to the extent to which personal traits, emotions, beliefs, and values are present (online and offline). Table 2 identifies the key traits and values that the interviewees considered to be important in relationships.

The importance of relationships was underlined by one interviewee who joined an international online glitch art community, from which a group of Mexican artists formed an online collective. It was in this collective that the artist started working with glitch to video, GIF, and 3D formats, to the point that they identify themselves as a 'glitch artist'. This is an example of how the adoption and interaction of explicit and tacit knowledge in this online community contributed to the identity formation of the artist's practice. As in an LED lamp, the work not only consists of a collection of technical knowledge, but permeates deeper into the artists' values and beliefs, also shaping knowledge processes in future collaborations:

Table 2

Important traits and values in relationships.

Trait / Value	Description
<i>Openness to share and receive knowledge/ Flexibility</i>	Being open and flexible to share and receive knowledge. Particularly relevant in tacit knowledge, where the distant perspectives and ideas are shared.
<i>Shared interests</i>	Having similar topics and explorations of interest.
<i>Reliability / Responsibility</i>	Particularly mentioned in the context of work teams and projects. Being part of a positive work environment and being held accountable to perform one's tasks in a timely manner.
<i>Respect</i>	Avoiding toxic judgments towards different opinions and perspectives. Valuing each other's time, work, processes.
<i>Trust</i>	Having honest intentions, a strong social tie that allows to rely on each other.
<i>Affection</i>	Being comfortable and having a good time together, often shown in friendly relationships. Being able to create affective bonds.
<i>Critical thinking</i>	Having reflective conversations and a critical approach towards the subject in question. Being objective and argumentative.
<i>Curiosity</i>	Showing eagerness to learn together. Being attentive in the conversation. Asking questions and wanting to know more.
<i>Empathy</i>	Learning by seeing each other's perspectives.
<i>Honesty</i>	To express the real intentions of the collaboration, project, and/or conversation.

It starts to escalate, first you see their work, then you talk a bit more with them (...); but it is simply a connection that exists between tastes and ways of working. (...) Collaborating has to do with talking about what we want to express. Those that work for me are the ones in which they send me something, I destroy it, send it back, they add other things, until we reach something. (Interviewee 2)

Collaborative processes are not very straightforward and don't always have a clear methodology (see example of artwork in Fig. 4). This is also why metaphors, such as the LED lamp, prove so valuable for sense-making as well as symbolic meaning.

These values and attitudes enable coping with the natural uncertainty of producing cultural goods (Potts et al., 2008; Scott, 2010): 'I believe that it is important to learn how to work, respect, and create from disagreements. We can disagree but be together, and maybe produce disagreements, instead of agreements' (Interviewee 12). The artists expressed the need to create spaces and situations where such exchanges can happen. The interactions between artists are fundamental to their career and identity (Scott, 2010, 2012), which corroborates Currid's (2007) idea that the social milieu is important for the creative economy to flourish. This involves the effort and commitment from all the participants to learn how to create such contexts, formally and informally, facilitated by an individual or by an institution:

When we organize talks, one of the ways in which we break down formal schemes is by providing coffee or other amenities to all the participants. These might appear as very simple gestures but in one way or the other they break the division between who is from the organization and who is an attendee. This allows for a different type of conversation and the reunion becomes much closer. (Interviewee 10).

It is in active participation where both explicit and tacit knowledge converge, to create new forms of collaboration, and thus new knowledge (Nonaka & Toyama, 2003). As in the quote above, spatial proximity between actors can be an important factor in promoting conversations and dialogues (Clare, 2013; Drake, 2003). Having shared interests, expressing honesty, respect, and commitment that enhances trust between actors, may compensate for the risks perceived in offline but also online relationships.

One of the interviewees explains how an ideology which they identify as 'audiovisual guerrilla' and technological appropriation, became more relevant after working with video art for a while. In the artist's view, this ideology considered the premise that anyone could do video art and could express personal narratives, contributing to the democratization of media. After creating an artistic collective with other like-minded artists, this idea was further developed in their artworks. Inspired by the political and social climate in his native city ca. 2005, they developed a project in which they mapped the surveillance cameras in the city, often placed in areas where the middle and 'bohemian' class would normally gather. After a couple of iterations using a printed format that was distributed in the city, this map was shared with the public digitally, supported by funding from one of the main multimedia institutions in the country and following a desire to 'recoup' the new technologies that were becoming more available but often used only by big corporations. The digital app would send a notification to its users when approaching a surveillance camera, allowing them to add other cameras as well. This project became relevant in Mexico and in other parts of the world, 'due to its capacity to connect with other places' (Interviewee 8). In a project such as this one, the exchange of tacit knowledge, with ideas, values, perspectives, and beliefs of the artists involved allowed a collaboration that transcended the mere act of 'developing an app' with a functional use to the population affected by certain policies (see example Fig. 5).

These findings are in line with the conceptualisation of a knowledge-creating place, i.e. 'the way of organizing that is based on the meaning it creates (...). (...) The boundary of Ba [knowledge-creating place] is fluid and can be changed quickly, as it is set by the participants' (Nonaka & Toyama, 2003, p. 104). Contrary to the organizational knowledge-creating place, in the case of this research, a knowledge-creating place becomes a space created and facilitated by individuals, which can be in formal and informal settings (see Fig. 6). These artists have their own individual contexts and they share a certain context motivated by their shared interests. This is also connected to the values and personality traits they have, which are part of the common existential knowledge-creating place. The physical spaces where these encounters and dialogues take place form the physical knowledge-creating place. These can be related to formal events (such as exhibitions, symposiums, conferences, workshops) or informal events (such as social gatherings, parties, concerts). The virtual knowledge-creating places, in which many exchanges and sharing happen, are highlighted by the interviewees as fundamental to knowledge creation.

All these elements form the holistic knowledge-creating place of new media artists. It is in this shared context, where creative tensions are allowed to emerge, that new meanings surface, and new knowledge is created, adopted and transferred, the end result not being necessarily the most significant. The processes of knowledge in virtual spaces implies much more than only 'generic information

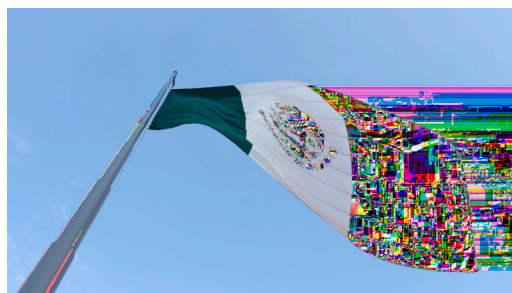


Fig. 4. Descomposición Nacional (2015) Martha Maya.

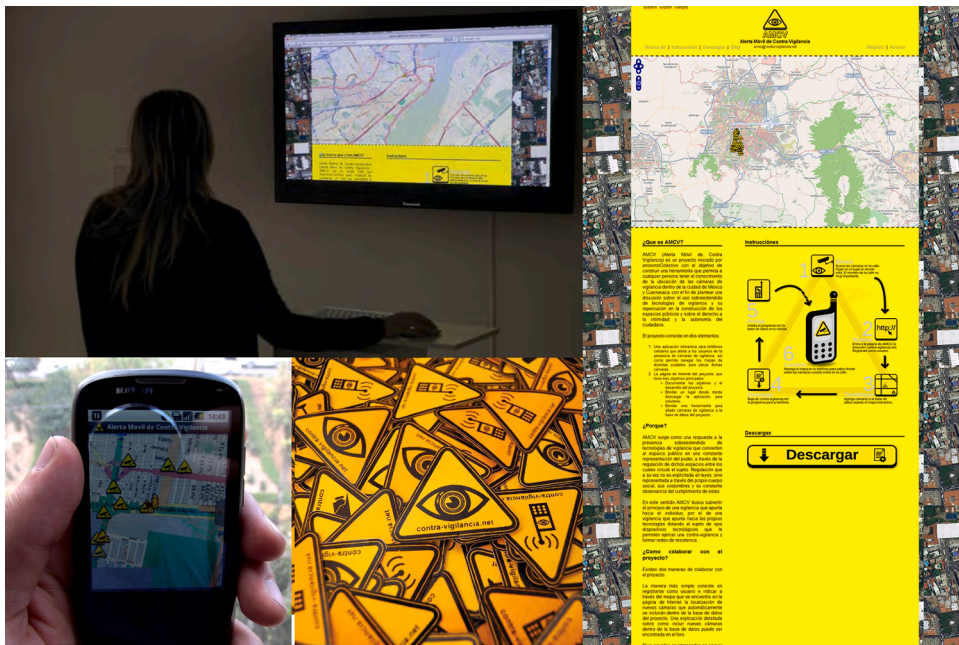


Fig. 5. AMCV (Alerta Movil de Contra-vigilancia).(2008–2012) Anónimo Colectivo.

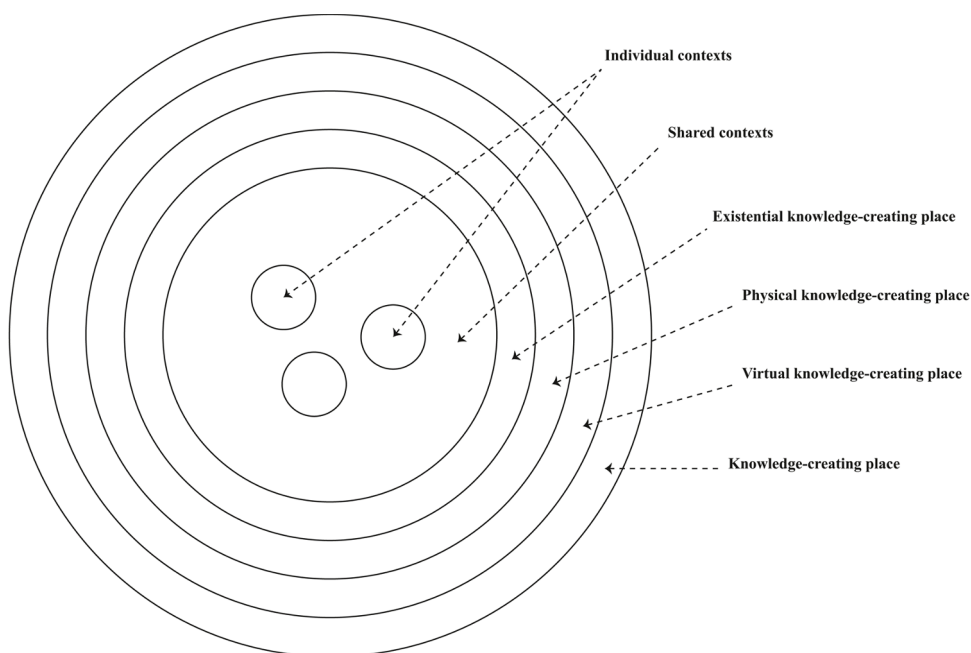


Fig. 6. The knowledge-creating place (adapted from Nonaka & Toyama, 2003).

skills and competences' (Robins, 1999: 20). Once again, explicit and tacit knowledge become intertwined and overlap in different contexts. In this sense, knowledge processes in the creation of new media art endeavours are very much contextual, and a result of a fertile knowledge-creating place (Ba). The LED lamp metaphor is the illustration of how artists look for and make complex and buzzing knowledge exchanges at different levels. This analysis of the Ba in artistic contexts suggests that this knowledge-creating place is central to artists' (collaborative) creative process.

9. Final remarks

The circumstances in which the creation, transfer, and adoption of knowledge can be fostered in the context of new media art are varied, and explicit and tacit knowledge are intertwined. New media artists are driven by curiosity and openness to different fields, and that is why interdisciplinarity characterises their artistic endeavours. There are formal moments where their activity occurs, while informal spaces enable fluid knowledge exchanges. By analysing the knowledge flows – knowledge as a solution, as experience, and as socially created (Snider & Nissen, 2003) – it becomes clear that the technical knowledge is only a small part of the overall knowledge exchange. New media art is a field where artistic, scientific and technological research come together to give shape to a particular ‘form of reflection’ (Brouwer et al., 2012, p. 7). Such research and reflection are closely connected to learning, which cuts across the different types of knowledge flows.

The way in which artists learn and interact are related to their personality; and values such as openness, curiosity, respect or trust shape their relationships (Robins, 1999) and thus the context for a shared interest to become the platform for knowledge creation, adoption, and exchange. In its different layers, the holistic concept of Ba as a knowledge-creating place (Nonaka & Toyama, 2003) contributes to understanding the fluid contexts in which meaning creation occurs. Using such a concept in the framework of artistic creation opens the way to understanding creativity and creative processes as open exchanges where creative tensions, dialogues and synergies become the backbone of a quest, a creation and a continuous learning process. This knowledge-creating place can occur physically and virtually, in safe environments where trust is a basis for positive social relations and creative collaborations.

The present exploratory study serves the purpose of providing insights on contextual factors in this ever-changing field, opening up avenues for new questions which remain largely unanswered. First, an ethnographic study, larger in scope and in timeframe, would allow for a deeper understanding of tacit knowledge processes, as it wouldn't rely on interviewees alone (which are based on declarative knowledge, Lizardo, 2017) and it could provide new insights in terms of context, as well as shifts and collaborative processes. Second, these collaborative processes in the context of new media art are still understudied and both the artistic communities and academia could use the basis of this research to continue investigating collaborative processes, in particular related to cultural homophily mechanisms (e.g. Basov, 2020). These could help in our understanding of the ongoing creation of this particular art world (Becker, 1982) with its knowledge exchanges among artists and also non-artists. Finally, because these exchanges are motivated and, to a certain extent, rely on technical knowledge, an Actor Network Theory approach could provide insights on the obstacles of producing new media art, such as technical issues or the continuous learning needed to make machines do new things.

Despite being limited in geographical scope to Mexico City, this research provides new information and understanding about new media art by looking into social phenomena that affect the production of artworks. The obstacles and context of the production of new media art in Mexico City seem to corroborate the enthusiasm, energy, and the exceptional curiosity and trust that new media artists are known for. Their shared interests (Collins, 2004) lead to them to create, adopt and transfer knowledge through collaborative processes in projects where the end result is often unknown. In this sense, the LED lamp metaphor, used widely in this circle in Mexico City, conveys a message and illustrates how the knowledge exchanges and collaborative interactions are perceived: explicit and tacit knowledge are present, they flow in different directions and nurture each other in a creative process that is not linear and is fluid according to contextual factors. In this regard, the results of this study provide relevant information about the way in which learning processes happen between new media artists and the actors in their social networks. The findings can be helpful for actors in the cultural and creative industries who play a role in facilitating creative and knowledge processes and flows of (new media) artists.

The framework suggested in this study can also be useful not only to understand knowledge management as a collaborative process among (new media) artists, but also to enrich the understanding of contextual factors in knowledge management processes which can also be relevant for organizations who want to improve the flow of creative-related knowledge.

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Declaration of Competing Interest

None.

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